

STEP RESPONSE UNIT - 8 OR DC Transients

Discuss the effects of switching on DC supply connected to RL & RC circuits

STEP RESPONSE OF RL SERIES CIRCUIT

Consider a coil (inductor L) in series with Resistance R shown in figure 1. connected across source (dc source) through switch S .

When ckt. is closed through switch S , the current increases gradually and takes some time to attain final value.

The reason for the delay is that when current increases to attain final value, an emf. (self-induced e.m.f) is induced in inductor which opposes the change in current.

Let current through inductor (L) and resistor (R) be i Amperes. and increases at rate of (di/dt) amperes.

Then applied voltage is :-

$$V = (iR + v_L) = V_R + v_L \quad [\text{Applying Kirchhoff voltage law on loop ABCD}]$$

where $v_R \rightarrow$ voltage across resistance (R)

$v_L \rightarrow$ voltage across inductor (L)

Case A: $\rightarrow$

RISE OF CURRENT IN INDUCTIVE CIRCUIT :-

$$V = (iR + L \frac{di}{dt}) \quad [v_L = -e_L = L \frac{di}{dt}] \quad [e_L = -L \frac{di}{dt}]$$

Reshuffling the terms,

$$(V - iR) = L \left(\frac{di}{dt} \right)$$

$$\text{or} \quad \left(\frac{dt}{L} \right) = \left[\frac{di}{(V - iR)} \right]$$

Multiplying both sides by $(-R)$ and then integrating both sides,

$$\int \frac{-R \cdot di}{(V - iR)} = \frac{-R}{L} \int 1 dt$$

$$\log_e (V - iR) = \left(\frac{-R}{L} t + K \right) \quad \text{--- (1)}$$

$\rightarrow$ constant of integration

Value of K can be determined from initial conditions i.e. at $t=0$ and $i=0$. Substituting these values in eqn 1

$$\log_e (V) = K$$

Now equation 1 becomes,

$$\log_e (V - iR) = \left[\frac{-R}{L} t + \log_e (V) \right] \quad \rightarrow \text{by substituting the value of } K$$

$$\log_e (V - iR) - \log_e (V) = \left(\frac{-R \cdot t}{L} \right)$$

$$\log_e \left(\frac{V - iR}{V} \right) = \left(\frac{-R \cdot t}{L} \right)$$

changing the ^{above} equation to exponent form;

$$\left(\frac{V - iR}{V} \right) = \left(e^{-Rt/L} \right)$$

$$(V - iR) = V \cdot e^{-Rt/L}$$

$$(V - iR) = V e^{-Rt/L}$$

$$(V - V e^{-Rt/L}) = iR$$

$$V (1 - e^{-Rt/L}) = iR$$

$$iR = V (1 - e^{-Rt/L})$$

$$i = \left(\frac{V}{R} \right) (1 - e^{-Rt/L})$$

$$i = \frac{V}{R} [1 - e^{-Rt/L}]$$

$\left(\frac{V}{R} \right)$ is the final value of steady current, I attained by ckt.

$$\therefore i = I (1 - e^{-Rt/L})$$

$$(2) \quad i = I (1 - e^{-Rt/L})$$

The above expression shows the rise of current w.r.t. time.

The exponent of e is (Rt/L)

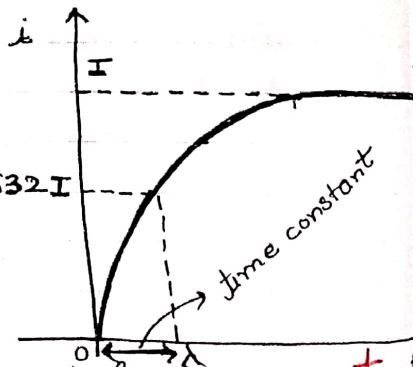
The quantity $\left(\frac{L}{R} \right)$ is called the 'time constant' of circuit and it affects the rise of current in circuit.

Time constant is represented by ' λ '.

$$\therefore \text{Time constant, } \lambda = \left(\frac{L}{R} \right)$$

So equation (2) can be represented as: $0.632I$

$$i = I (1 - e^{-t/\lambda})$$



The current waveform (rise of current) is exponential as shown in figure 2.

At time, $t = \left(\frac{L}{R} \right)$ is at Time = time constant (λ), the current is

$$i = I (1 - e^{-1}) = 0.632I$$

$$\text{or } i = 0.632I$$

$\therefore$ time constant of RL circuit is the time required for current in RL series circuit, to reach 0.632 times its steady state value (I)

Note - Unit of time constant is seconds.

Case B: →

Case II)

DECAY OF CURRENT IN ^(RL) INDUCTIVE CIRCUIT: -

Now if the switch is immediately opened and thrown to position 2 when RL series ckt is conducting, then the current will not decay to zero immediately because of energy stored across inductor in its magnetic field. (Figure 3)

Let. at any instant, the current in circuit be i amperes and is decreasing at rate of (di/dt) .

Applying KVL in loop ABCD;

$$0 = iR + L \left(\frac{di}{dt} \right) \Rightarrow (-iR) = L \left(\frac{di}{dt} \right)$$

$$\left(\frac{di}{i} \right) = \left(-\frac{R}{L} dt \right)$$

Integrating both sides, we get

$$\int \frac{di}{i} = \int \left(-\frac{R}{L} \right) dt$$

$$\log_e i = \left(-\frac{R}{L} t + K \right) \quad \text{--- (III)}$$

const. of integration.

value of K can be determined from initial conditions; i.e. $t=0$, $i=I$

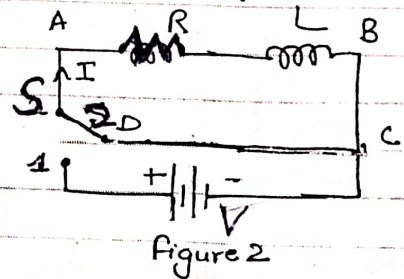
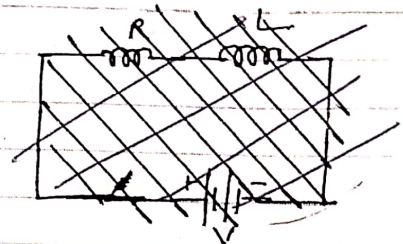
$$\log_e I = (0 + K) \quad \text{or} \quad K = \log_e I$$

Substituting value of K in eqn III.

$$\log_e i = \left(-\frac{R}{L} t \right) + \log_e I$$

$$\log_e i - \log_e I = \left(-\frac{R}{L} t \right)$$

$$\log_e \left(\frac{i}{I} \right) = \left(-\frac{R}{L} t \right)$$



(14/10/20)

changing the equation to exponent form;

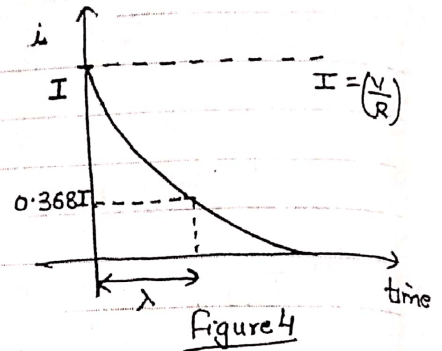
$$\left(\frac{i}{I}\right) = \left(e^{-Rt/L}\right)$$

or $i = I \left(e^{-Rt/L}\right)$

Again substituting $\tau \left(\frac{L}{R}\right)$ as time constant (λ) in above eqn.

$$i = I \cdot e^{-t/\lambda} \quad - (4)$$

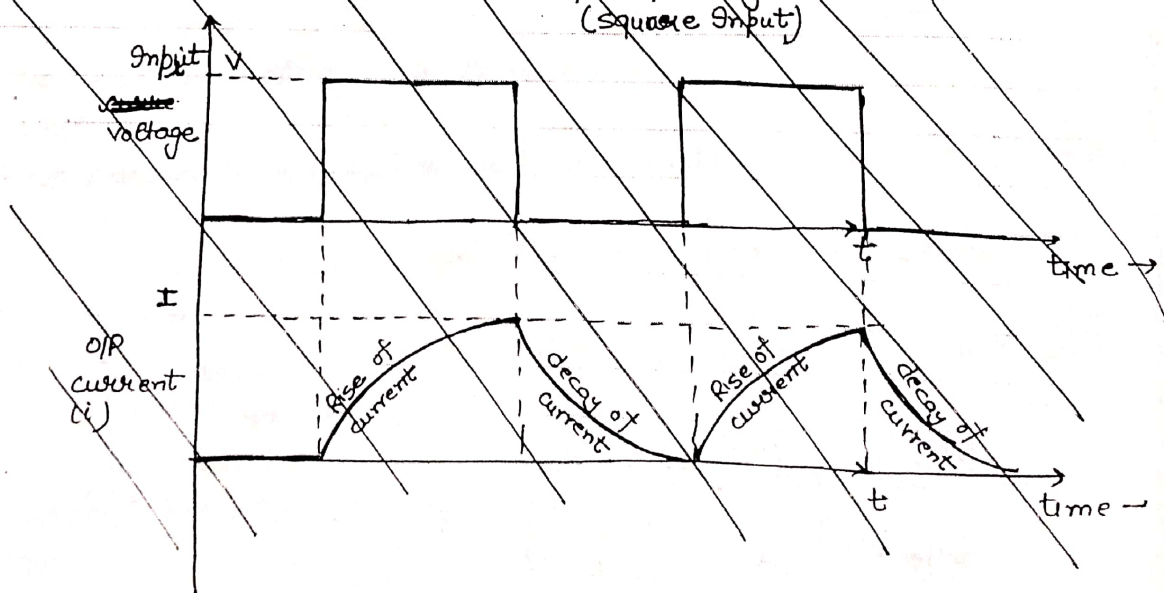
Plots of i wrt t is shown in figure 4.
In above equation 4, substituting $t = \lambda$
i.e. at time = time constant (λ),



$$i = I \left(e^{-1}\right) \Rightarrow i = 0.368I$$

So, time constant can also be defined as time required by current in RL series circuit to decay to 0.368 times of the steady state value (I) is called time constant of circuit.

- Response of RL series circuit to step input signal: -
(square input)

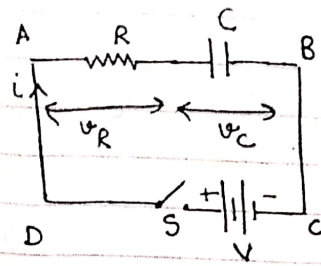


Gkambr Sidhu

STEP RESPONSE OF RC SERIES CIRCUIT:-

Case I: →
Rise of voltage in RC circuit:- or

Charging of Capacitor



- Consider a capacitor connected across Resistor (R) through switch (S)
- The voltage across capacitor will not rise instantaneously when switch (S) is closed instantaneously.
- The rate of rise of voltage across RC circuit is calculated as below:-

The current through Resistor (i) = current through capacitor (i_c)

$$\left(\frac{v_R}{R}\right) = C \left(\frac{dv_C}{dt}\right) \quad \text{--- (1)}$$

[∵ in series, current remains same]

Apply KVL in loop ABCD

$$(v_R + v_C) = V \Rightarrow v_R = (V - v_C)$$

Substituting the value of v_R in equation (1)

$$\left(\frac{V - v_C}{R}\right) = C \left(\frac{dv_C}{dt}\right) \quad \Rightarrow \quad \frac{V - v_C}{R} = C \left(\frac{dv_C}{dt}\right)$$

$$(V - v_C) = RC \left(\frac{dv_C}{dt}\right)$$

Reshuffling the terms;

$$dt = RC \left(\frac{dv_C}{V - v_C}\right)$$

Integrating both sides,

$$\int dt = RC \int \frac{dv_C}{(V - v_C)}$$

$$t = -RC \log_e (V - v_C) + K \quad \text{--- (2)}$$

↳ constant of proportionality

The value of K can be determined from initial conditions i.e. -
at $t=0$ and $v_C=0$. Substituting these values in eqn 2.

$$0 = -RC \log_e (V - 0) + K$$

$$K = RC \log_e (V)$$

Substituting value of K in eqn (2)

$$t = -RC \log_e (V - v_C) + RC \log_e (V)$$

RIK

$$t = RC [e^{\log_e V} - e^{\log_e (V - v_c)}]$$

$$t = RC \left[e^{\log_e \left(\frac{V}{V - v_c} \right)} \right] \Rightarrow \left(\frac{t}{RC} \right) = \log_e \left(\frac{V}{V - v_c} \right)$$

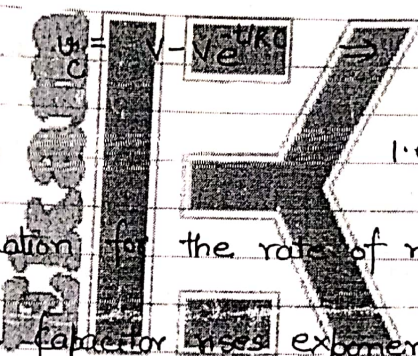
Changing to exponent form;

$$\left(\frac{t}{RC} \right) = \log_e \left(\frac{V}{V - v_c} \right)$$

Changing the above eqⁿ to exponent form;

$$e^{-t/RC} = \left(\frac{V - v_c}{V} \right)$$

$$V \cdot e^{-t/RC} = V - v_c$$



$$v_c = V(1 - e^{-t/RC})$$

$$\text{i.e. } v_c = V(1 - e^{-t/RC}) \quad - (3)$$

- Above is the equation for the rate of rise of voltage across capacitor.

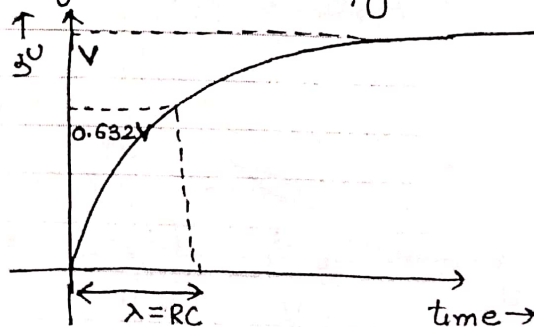
- The voltage across capacitor rises exponentially as shown in figure 2.

- The term 'RC' is actually time constant (λ) of the RC circuit.

$$\text{i.e. } \lambda = RC$$

Substituting $\lambda = RC$ in eqn (3)

$$v_c = V(1 - e^{-t/\lambda}) \quad - (4)$$



- At time, $t = \lambda$ i.e. at $t = RC$; $v_c = V(1 - e^{-1}) = V(1 - e^{-1})$

$$v_c = 0.632V$$

$\therefore$ time constant of RC series circuit is the ~~rate at which~~ ^{time} time required for the voltage across RC series circuit to reach 0.632 times its steady state value (V).

se II) DECAY OF ~~CURRENT~~ ^{VOLTAGE} ACROSS RC SERIES CIRCUIT :-

Discharge of a Capacitor

Consider a RC series circuit in which capacitor is initially charged to voltage (V_0) and the switch (S) is thrown to position 2 as shown in figure 3.

The figure 3 can be represented by figure 4. equivalently

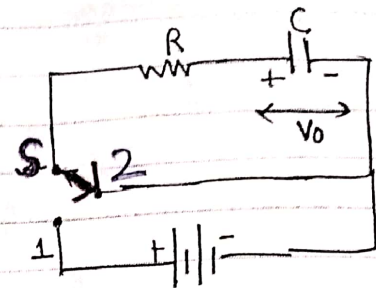
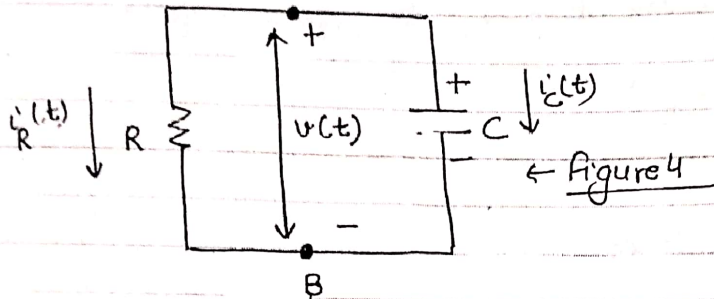


Figure 3
battery is cut off

The capacitor C now discharge through resistor (R) and the voltage across capacitor will vary. Let the varying voltage be represented as $v(t)$ across AB terminals.

- The currents flowing through Resistor and capacitor are shown with the sourcing terminal A.

- Applying KCL at point A; $i_R(t) + i_C(t) = 0$

$$C \frac{dv(t)}{dt} + \frac{v(t)}{R} = 0$$

$$\left(\frac{dv(t)}{v} \right) = - \frac{1}{RC} dt$$

$$\int \frac{dv(t)}{v} = - \frac{1}{RC} \int dt$$

$$\log_e v(t) = \left[-\frac{1}{RC} t + K \right] \quad \text{--- (1)}$$

constant of integration

Initially at $t=0$, $v(t) = V_0$. [initial voltage across capacitor]
 $\log_e V_0 = K$

Substituting value of K in eqn (1)

$$\log_e v(t) = \left[\left(-\frac{t}{RC} \right) + \log_e V_0 \right]$$

$$\log_e v(t) - \log_e V_0 = \left(\frac{-t}{RC} \right)$$

$$\log_e \left[\frac{v(t)}{V_0} \right] = \left(\frac{-t}{RC} \right)$$

Changing the eqn to exponent form,

$$\left[\frac{v(t)}{V_0} \right] = \left[e^{-t/RC} \right]$$

$$v(t) = V_0 \cdot e^{-t/RC} \rightarrow \text{eqn for decay for voltage in RC ckt.}$$

Substituting time const., $\lambda = RC$

$$v(t) = V_0 \cdot e^{-t/\lambda}$$

- The ~~current~~ ^{Voltage} through capacitor decays exponentially as shown in figure 4.

- At time, $t = \lambda$ or $t = RC$ seconds

$$v(t) = V_0 \cdot e^{-\lambda/\lambda} = V_0 \cdot e^{-1} = V_0 \cdot 0.368$$

$$v(t) = 0.368 V_0$$

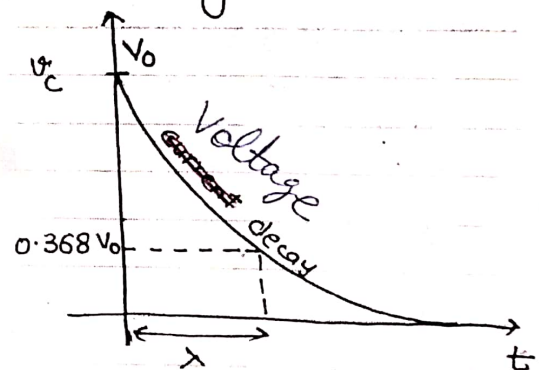
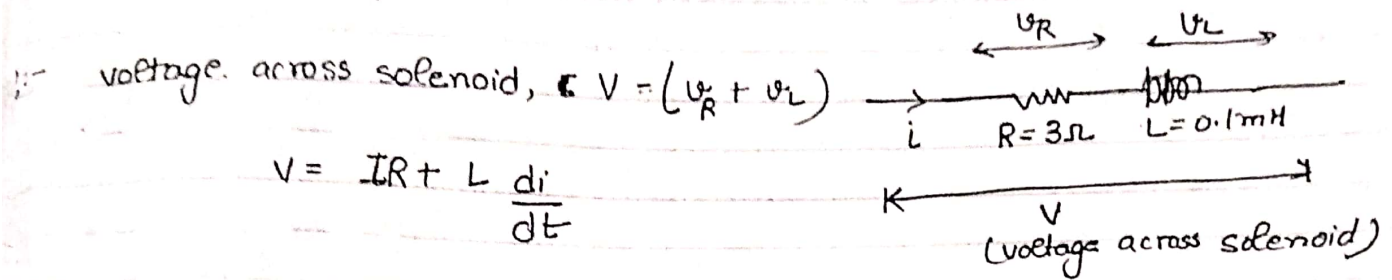


Figure 4

$\therefore$ time const. for RC series ckt can be defined as the time taken by ~~current~~ ^{voltage} in RC series ckt to decay to 0.368 times its initial value (V_0).

Ekambir Sidhu

- 1) The resistance and inductance of coil are 3Ω and 0.1mH respectively. What is the potential across the ends of solenoid at instant when the current is 1A , but increasing at rate of 1000A per second ?



Given; $I = 1\text{A}$; $R = 3\Omega$; $L = 0.1 \times 10^{-3}\text{H}$ and $\left(\frac{di}{dt}\right) = 1000\text{A/s}$

$$\therefore E = (1 \times 3) + (0.1 \times 10^{-3} \times 1000)$$

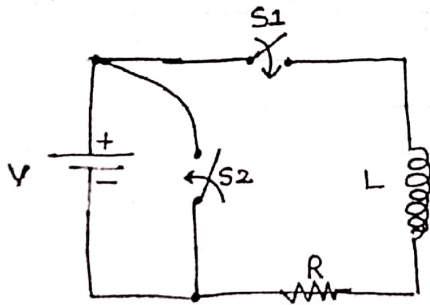
$$E = (3 + 0.1)$$

$$E = 3.1\text{ volts}$$

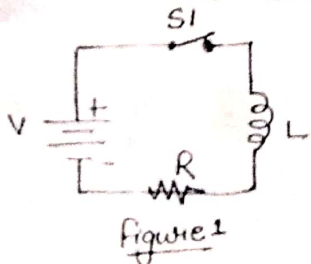
Answer

- 1) Consider series RL circuit connected to battery and switch as shown in figure below. Switch closes at $t = 2\text{ seconds}$; switch S_2 closes at $t = 5\text{ seconds}$.

- (a) Find the current $i(t)$ in resistance R ;
(b) Plot the current $i(t)$ for $0 \leq t \leq 10\text{ seconds}$ to the scale.
Assume $\left(\frac{R}{L}\right)$ ratio equal to 100



Solution: (a) When the switch S_1 is closed and S_2 remains open, the circuit becomes as shown in figure. 1 below

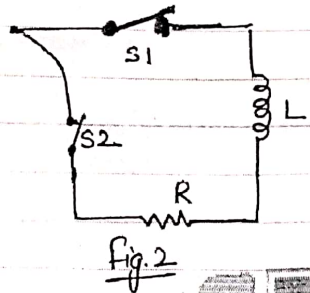


The switch S_1 is closed after 2 seconds, therefore upto 2 seconds, no current flows upto 2 seconds. (shown by 0A in figure 3.)

- After 2 seconds, the switch S_1 is closed, current starts rising exponentially (shown by part ~~AB~~ of curve - figure 3) given by equation:

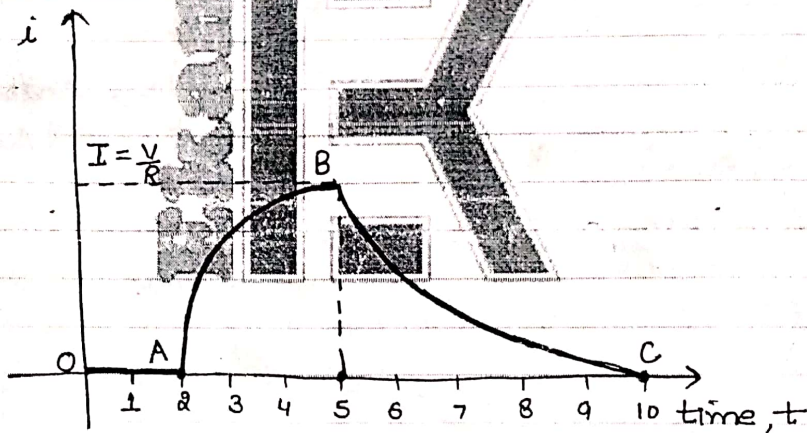
$$i(t) = \frac{V}{R} (1 - e^{-Rt/L}) = \left(\frac{V}{R}\right) [1 - e^{-100t}]$$

- After $t = 5$ seconds, S_2 is ~~is~~ closed as shown in fig. 2 below:



- The current starts decaying exponentially and is given by equation (shown by curve BC in fig. 3)

$$i(t) = \frac{V}{R} (e^{-Rt/L}) = \frac{V}{R} (e^{-100t})$$



- Q3) A coil having $L = 6.4 \text{ H}$ and $R = 8 \Omega$ is connected to constant 200 V supply source. How long does it take the voltage across reach 100 V ?

Solution :-

$$i = \frac{V}{R} (1 - e^{-t/\lambda})$$

$$iR = V (1 - e^{-t/\lambda})$$

Given ; $iR = 100 \text{ V}$; $V = 200 \text{ volts}$, $\lambda = \left(\frac{L}{R}\right) = \left(\frac{6.4}{8}\right) = 0.8$.

Substituting the values ; $100 = 200 (1 - e^{-t/0.8})$

$$0.5 (1 - e^{-t/0.8})$$

$$e^{-t/0.8} = 0.5 \Rightarrow e^{t/0.8} = \left(\frac{1}{0.5}\right) \Rightarrow \left(\frac{t}{0.8}\right) \log e = \log 2$$

$$t = (0.8 \times 0.6931) = 0.5545 \text{ seconds And}$$